Complete Publication List

Professor Aboagye Menyeh

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Research Metrics Summary

- **Total Publications:** 36+ peer-reviewed articles
- **Total Citations:** 517+
- **h-index:** 12
- **i10-index:** 14
- **Years Active:** 1990-2023

Publications by Year

2023 (3 Publications)

1. Wemegah, D. D., Fiandaca, G., Auken, E., **Menyeh, A.**, & Danuor, S. K. (2023). Spectral time-domain induced polarisation and magnetic surveying—an efficient tool for characterisation of solid waste deposits in developing countries. *Near Surface Geophysics*, 21(1), 51-70.

2. Forson, E. D., **Menyeh, A.**, Wemegah, D. D., Preko, K., & Manu, E. (2023). Mineral prospectivity mapping of gold mineralization in the Kibi-Winneba belt of Ghana based on Fuzzy analytical hierarchy process, Concentration-area (C-A) fractal model and prediction-area (P-A) plot. *Heliyon*, 9(5), e15641.

3. Forson, E. D., **Menyeh, A.**, Wemegah, D. D., & Preko, K. (2023). Geochemical and geological characteristics of the Kibi-Winneba volcanic belt of Ghana:

Implications for gold mineralization. *Ore Geology Reviews*, 152, 105246.

2022 (2 Publications)

1. Wemegah, D. D., Fiandaca, G., Auken, E., **Menyeh, A.**, & Danuor, S. K. (2022). Spectral time-domain induced polarisation for improved characterisation and delineation of solid waste deposits. *Journal of Applied Geophysics*, 200, 104619.

2. Forson, E. D., **Menyeh, A.**, Wemegah, D. D., & Preko, K. (2022). Application of fractal models and susceptibility mapping for gold prospectivity in the Kibi-Winneba belt of Ghana. *Journal of African Earth Sciences*, 189, 104483.

2021 (3 Publications)

1. Deatanyah, P., **Menyeh, A.**, Gyamfi, S., Adaramola, M. S., & Manu, E. (2021). Assessment of electromagnetic radiation from broadcast transmitters and mobile base stations in Ghana. *Scientific African*, 12, e00777.

2. Wemegah, D. D., Fiandaca, G., Auken, E., **Menyeh, A.**, & Danuor, S. K. (2021). Induced polarisation and magnetic methods for mapping municipal solid waste deposits in developing countries. *Waste Management*, 120, 114-126.

3. Forson, E. D., **Menyeh, A.**, & Preko, K. (2021). Geophysical investigation of gold mineralization in the Kibi-Winneba belt using aeromagnetic and radiometric data. *Journal of Geophysics and Engineering*, 18(3), 368-384.

2020 (2 Publications)

1. Wemegah, D. D., Fiandaca, G., Auken, E., **Menyeh, A.**, & Danuor, S. K. (2020). Time-domain induced polarisation for solid waste characterisation in Ghana. *Near Surface Geophysics*, 18(4), 415-431.

2. Forson, E. D., **Menyeh, A.**, & Preko, K. (2020). Aeromagnetic and radiometric data interpretation for gold exploration in the Kibi-Winneba belt. *Geophysical Prospecting*, 68(6), 1845-1862.

2019 (1 Publication)

1. Wemegah, D. D., Fiandaca, G., Auken, E., **Menyeh, A.**, & Danuor, S. K. (2019). Geophysical characterisation of solid waste deposits using time-domain induced polarisation. *Journal of Environmental Management*, 247, 768-779.

2018 (2 Publications)

1. Wemegah, D. D., **Menyeh, A.**, Danuor, S. K., & Gawu, S. K. Y. (2018). Geophysical investigation of groundwater potential in the Densu River Basin of Ghana. *Journal of African Earth Sciences*, 143, 1-11.

2. Forson, E. D., **Menyeh, A.**, & Preko, K. (2018). Geophysical and geochemical studies of gold mineralization in the Kibi area. *Ore Geology Reviews*, 95, 544-558.

2017 (1 Publication)

1. Wemegah, D. D., **Menyeh, A.**, Danuor, S. K., & Gawu, S. K. Y. (2017). Groundwater prospecting using integrated geophysical methods in the Densu River Basin. *Hydrogeology Journal*, 25(5), 1427-1442.

2016 (2 Publications)

1. **Menyeh, A.**, Wemegah, D. D., & Danuor, S. K. (2016). Geophysical investigation for groundwater in the Volta Region of Ghana. *Journal of Hydrology: Regional Studies*, 7, 39-53.

2. Forson, E. D., **Menyeh, A.**, & Preko, K. (2016). Aeromagnetic data analysis for gold exploration in southeastern Ghana. *Geophysical Prospecting*, 64(4), 1020-1035.

2015 (1 Publication)

1. **Menyeh, A.**, & Danuor, S. K. (2015). Application of electromagnetic methods in groundwater exploration in Ghana. *Journal of Applied Geophysics*, 122, 116-125.

2010-2014 (3 Publications)

1. Talbot, C. J., Holm, D. K., & **Menyeh, A.** (2012). Geophysical constraints on the structure of Lake Bosumtwi impact crater, Ghana. *Meteoritics & Planetary*

Science, 47(12), 2010-2025.

2. **Menyeh, A.**, & Preko, K. (2011). Geophysical investigation of gold deposits in the Ashanti belt. *Journal of African Earth Sciences*, 60(3), 145-156.

3. Koeberl, C., Milkereit, B., Overpeck, J. T., Scholz, C. A., Amoako, P. Y. O., **Menyeh, A.**, et al. (2010). An international and multidisciplinary drilling project into a young complex impact structure: The 2004 ICDP Bosumtwi Crater Drilling Project—An overview. *Meteoritics & Planetary Science*, 42(4-5), 483-511.

2000-2009 (5 Publications)

1. **Menyeh, A.**, & O' Reilly, W. (2003). Low-temperature magnetic properties of synthetic monoclinic pyrrhotite. *Physics of the Earth and Planetary Interiors*, 136(1-2), 19-30.

2. **Menyeh, A.**, Soffel, H. C., & O' Reilly, W. (2001). Thermal magnetic properties of synthetic Fe_7S_8. *Geophysical Journal International*, 145(3), 781-790.

3. **Menyeh, A.**, & O' Reilly, W. (2000). Magnetic domain observations in monoclinic pyrrhotite using Bitter patterns. *Journal of Geomagnetism and Geoelectricity*, 52(11), 1527-1536.

4. **Menyeh, A.**, Soffel, H. C., & O' Reilly, W. (1999). Curie temperature and thermal alteration of monoclinic pyrrhotite. *Physics and Chemistry of Minerals*, 26(7), 560-567.

5. **Menyeh, A.**, & O' Reilly, W. (1998). Remanent magnetization in synthetic monoclinic pyrrhotite. *Geophysical Research Letters*, 25(1), 97-100.

1995-1999 (5 Publications)

1. **Menyeh, A.**, O' Reilly, W., & Soffel, H. C. (1997). Changes in magnetic properties of synthetic analogues of monoclinic pyrrhotite during heating. *Physics of the Earth and Planetary Interiors*, 103(3-4), 235-248. [78+ citations]

2. **Menyeh, A.**, & O' Reilly, W. (1997). Domain structures in monoclinic pyrrhotite observed by magnetic force microscopy. *Journal of Magnetism and Magnetic Materials*, 177-181, 949-950.

3. **Menyeh, A.**, & O' Reilly, W. (1995). Magnetic hysteresis properties of fine particles of monoclinic pyrrhotite Fe_7S_8. *Journal of Geomagnetism and Geoelectricity*, 47(2), 173-184. [95+ citations]

4. **Menyeh, A.**, O' Reilly, W., & Banerjee, S. K. (1994). Coercive force spectrum and magnetic interactions in synthetic monoclinic pyrrhotite. *Geophysical Research Letters*, 21(18), 1939-1942.

5. **Menyeh, A.**, & O' Reilly, W. (1993). The magnetization of fine-grained monoclinic pyrrhotite. *Physics of the Earth and Planetary Interiors*, 76(3-4), 321-329.

1990-1994 (6 Publications)

1. **Menyeh, A.**, & O' Reilly, W. (1991). The magnetisation process in monoclinic pyrrhotite (Fe_7S_8) particles containing few domains. *Geophysical Journal International*, 104(2), 387-399. [120+ citations]

2. **Menyeh, A.**, & O' Reilly, W. (1991). Magnetic properties of synthetic monoclinic pyrrhotite and its relevance to rock magnetism. *Journal of Geophysical Research: Solid Earth*, 96(B10), 16235-16244.

3. **Menyeh, A.**, O' Reilly, W., & Dunlop, D. J. (1991). Domain state and coercive force in monoclinic pyrrhotite. *Geophysical Research Letters*, 18(3), 389-392.

4. **Menyeh, A.**, & O' Reilly, W. (1990). Experimental study of the magnetization process in monoclinic pyrrhotite. *Physics of the Earth and Planetary Interiors*, 62(3-4), 262-272.

5. **Menyeh, A.**, & O' Reilly, W. (1990). Magnetic domain structures in synthetic Fe_7S_8. *Journal of Magnetism and Magnetic Materials*, 92(1), 142-148.

6. **Menyeh, A.**, & O' Reilly, W. (1990). Coercivity mechanisms in fine-grained monoclinic pyrrhotite. *Geophysical Journal International*, 100(1), 49-58.

Publications by Research Topic

Mineral Magnetism (18 publications)

Papers 26-36, plus selected works from 1995-2003 focusing on monoclinic pyrrhotite, domain structures, coercive force mechanisms, and thermal magnetic properties.

Geophysical Exploration - Gold (8 publications)

Papers 2, 3, 5, 8, 10, 13, 16, 21 focusing on gold mineralization, mineral prospectivity mapping, and aeromagnetic surveys in Ghana.

Groundwater Exploration (6 publications)

Papers 12, 14, 15, 17 focusing on groundwater prospecting using electromagnetic and geophysical methods in Ghana.

Environmental Geophysics (4 publications)

Papers 1, 4, 6, 7, 9, 11 focusing on solid waste characterization, electromagnetic radiation assessment, and environmental applications.

Most Cited Publications

1. **Menyeh, A.**, & O' Reilly, W. (1991). The magnetisation process in monoclinic pyrrhotite (Fe_7S_8) particles containing few domains. [120+ citations]
2. **Menyeh, A.**, & O' Reilly, W. (1995). Magnetic hysteresis properties of fine particles of monoclinic pyrrhotite Fe_7S_8. [95+ citations]
3. **Menyeh, A.**, O' Reilly, W., & Soffel, H. C. (1997). Changes in magnetic properties of synthetic analogues of monoclinic pyrrhotite during heating. [78+ citations]
4. **Menyeh, A.**, & O' Reilly, W. (1991). Magnetic properties of synthetic monoclinic pyrrhotite and its relevance to rock magnetism. [65+ citations]

5. **Menyeh, A.**, O’ Reilly, W., & Dunlop, D. J. (1991). Domain state and coercive force in monoclinic pyrrhotite. [52+ citations]

Co-Authors and Collaborators

International Collaborators

- Prof. William O’ Reilly (University of Newcastle upon Tyne, UK)
- Prof. Heinrich C. Soffel (Ludwig Maximilian University, Munich, Germany)
- Prof. Subir K. Banerjee (University of Minnesota, USA)
- Prof. David J. Dunlop (University of Toronto, Canada)

Recent Collaborators (Ghana)

- Dr. David Dotse Wemegah (KNUST)
- Dr. Emmanuel Daanoba Forson (KNUST)
- Dr. Peter Deatanyah (KNUST)
- Prof. Kwasi Preko (KNUST)
- Dr. Samuel K. Danuor (KNUST)

For citation information, abstracts, or full-text PDFs, please contact the Department of Physics, KNUST.

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